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B.Tech. Degree II Semester Regular Examination July 2024

CE/ME/SE 23-200-0201A COMPUTER PROGRAMMING AND PROBLEM SOLVING (2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

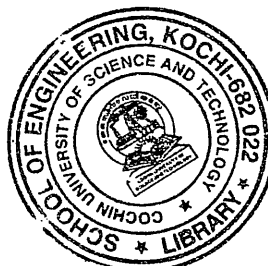
- CO1: Elucidate the basic architecture and functionalities of a computer and also recognize the hardware parts.
- CO2: Apply programming constructs of C language to solve real-world problems.
- CO3: Explore user-defined data structures like arrays in implementing solutions to problems like searching and sorting.
- CO4: Explore user-defined data structures like structures, unions and pointers in implementing solutions.
- CO5: Design and Develop Solutions to problems using modular programming constructs using functions.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
- PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a) What are the key differences between hardware and software and how do they interact with each other in a computer system? Provide examples of how hardware and software components are integrated in a typical computer system.	2	2	L4	1	1
(b) Explain the function of (i) Compiler (ii) Interpreter (iii) Loader (iv) Linker.	2	2	L2	1	1
(c) What is the output of the following program when input is 5? #include <stdio.h> int main() { int num; scanf("%d", & num); int i, j, count = 0; for(i=1; i<=num; i++){ for(j=1; j<=num; j++){ count++; } } printf("%d", count); return 0; }	2	2	L4	2	1
A) 25 B) 20 C) 15 D) 10					
(d) Explain Parameter Passing Mechanism in C programming.	2	2	L6	5	3
(e) What are the primary file handling operations in C?	2	2	L3	5	5

(P.T.O.)



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PART B

		(4 × 10 = 40)	Marks	BL	CO	PO
II.	(a) Differentiate between various looping constructs (for, while, do while) in C program with examples.		6	L2	2	1
	(b) Write a C program to find factorial of a number.					
	OR		4	L2	2	1
III.	(a) Write a C program to check whether the given number is armstrong or not.		6	L2	2	1
	(b) Explain the different data types in 'C' language.		4	L1	2	1
IV.	(a) Explain the different categories of software.		4	L4	1	1
	(b) Describe the functional components and operational stages of a digital computer, including input, processing, memory, control and output. Explain how these components interact to execute software instructions and perform computational tasks.		6	L2	1	3
	OR					
V.	(a) Differentiate between algorithm and flow chart and also write algorithm and flow chart for checking whether the given number is palindrome or not.		5	L1	1	1
	(b) Explain the problem solving methodology.		5	L2	1	3
VI.	(a) Explain different string handling functions in C with examples.		5	L2	3	1
	(b) Write a C program to arrange the names in alphabetical order.		5	L3	3	3
	OR					
VII.	(a) Write a C program to perform Matrix addition.		6	L3	3	3
	(b) Differentiate between arrays and structure with suitable examples.		4	L2	3	1
VIII.	(a) Explain about pointers and also explain the syntax for Pointer declaration and initialization.		5	L1	4	1
	(b) Explain dynamic memory allocation in C.		5	L2	4	5
	OR					
IX.	(a) Explain about command line Arguments in C.		5	L6	4	3
	(b) Explain file pointer and also explain file handling functions on sequential and random access files.		5	L3	4	5

Bloom's Taxonomy Levels

L1 – 14%, L2 – 43%, L3 – 20%, L4 – 14%, L6 – 9%.
